
Data Structures & Algorithms (CSC-214)Credit Hrs: **(Lec: 3)**

Prerequisite: Object Oriented Programming (CSC-213)

Semester: **Spring 2025**Instructor: **Hamna Anwar**GClassroom Code: **7e5qgu5**

Course Objective

This course covers the design of fundamental data structures and the algorithms that are applied on them. It aims to explain the pros and cons of different aspects of the design of the data structures, and to introduce tools for analyzing the time and space complexity. The course will also provide thorough practical experience via implementation of data structures in C++.

Text Books

1. Data Structures and Algorithms in C++ by Michael T. Goodrich, Roberto Tamassia & David Mount 2nd Edition (2011) Jon Wiley & Sons
2. C++ Data Structures: A Laboratory Course by Stefan Brandle, Jonathan Geisler, James Roberge & David Whittington 3rd Edition (2009) Jones and Bartlett Publishers.

Program Learning Outcomes (PLOs)

1. Academic Education
2. Knowledge for Solving Computing Problems
3. Problem Analysis
4. Design/Development of Solutions
5. Modern Tool Usage
6. Individual and Team Work
7. Communication
8. Computing Professionalism and Society
9. Ethics
10. Life-long Learning

Course Learning Outcomes (CLOs)

1. **Understand, explain and analyze** fundamental data structures (arrays, linked lists, stacks, queues, trees, and hash tables) along with their time and space complexities.
2. **Design and implement** efficient algorithms for sorting, searching, and managing dynamic data using recursion, trees, and hash tables.
3. **Apply** data structures to solve real-world computing problems such as memory management, scheduling, and caching.
4. **Demonstrate** effective teamwork and communication when developing data structure-based solutions.

	PLO1	PLO2	PLO3	PLO4	PLO6	PLO7
CLO1	x	x				
CLO2			x	x		
CLO3				x		
CLO4					x	x

Course Outline and Schedule

Week	Topics	CLO Mapping
Week 1	Recap: Arrays, strings, pointers, object oriented programming (classes, templates)	CLO1
Week 2	Intuitive introduction to space and time complexity. Array searching algorithms	CLO1, CLO2
Week 3	Array sorting algorithms.	CLO1, CLO2
Week 4	Singly linked lists.	CLO1, CLO2
Week 5	Doubly linked lists.	CLO1, CLO2
Week 6	Circularly linked lists.	CLO1, CLO2
Week 7	Linked list applications.	CLO3
Week 8	Stacks (via linked lists and arrays). Stack applications.	CLO1, CLO2, CLO3
Week 9	Midterm Exam	
Week 10	Queues (via linked lists and circular arrays). Double ended queues. Queue applications.	CLO1, CLO2, CLO3
Week 11	Recursion.	CLO1, CLO2
Week 12	Binary search trees.	CLO1, CLO2
Week 13	Self balancing binary search trees (AVL trees, red-black trees)	CLO1, CLO2
Week 14	Hash tables with linear probing and chaining. Hash functions.	CLO1, CLO2
Week 15	Using the C++ STL.	CLO3
Week 16	Heaps.	CLO1, CLO2
Week 17	Introduction to graphs.	CLO1, CLO2

Week	Topics	CLO Mapping
Week 18	Final Exam Week	
	Exam Scripts Shown	
	Declaration of Result	

Grading Breakdown and Policy

Assessment Description	Weight %	CLO Mapping
Quizzes	15%	CLO1
Midterm	25%	CLO1, CLO2, CLO3
Project	20%	CLO1, CLO2, CLO3, CLO4
Final Exam	40%	CLO1, CLO2, CLO3
Total	100%	

GRADING TABLE ABSOLUTE SCALE if no. of students in Final Exam are up to 20, otherwise RELATIVE SCALE			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A-
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B-
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C-
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F